

Lab 1: Knowledge Is Power

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1 Introduction

Knowledge Is Power (KIP) is a proposed digital platform focused on improving legal literacy among young adults. This document introduces the motivation behind KIP, the characteristics needed for an effective solution, and the overall purpose of the proposed product. It also provides an overview of KIP's features, capabilities, and technical components.

1.1 Societal Problem

Young people ages 16–19 increasingly encounter laws involving driving, digital conduct, work, school, ownership, contracts, and safety without fully understanding their inner workings and the consequences that may ensue. As they gain more independence into adulthood, this lack of legal awareness can lead to poor decisions and serious consequences. Despite this, law-related education remains largely absent from K-12 curricula, leaving young adults underprepared for real-world legal situations. Additionally, teen drivers face higher safety risks that are often linked to poor judgement and decision-making. [4.1, 4.2, 4.3, 4.4]

1.2 Needed Solution Characteristics

The platform should teach legal concepts through interactive activities that ask users to apply laws to realistic situations rather than simply memorize definitions. To remain accessible, it should use clear and consistent language, concise explanations, mobile-friendly web design, and low-stakes practice games that students can complete independently. It should also explain the connection between each lesson and its official legal sources while introducing legal vocabulary in context so that users can recognize practical terms used across different legal topics.

1.3 Product Introduction

Knowledge Is Power (KIP) is a digital legal literacy platform designed to teach essential, everyday laws to young adults. It serves as a high-quality supplemental educational aid for parents, teachers, and school administrators. KIP's goal is to help young adults recognize legally significant situations and understand the potential consequences of their choices, to prepare them to make more informed decisions. The prototype will demonstrate these goals through a web-based service that supports user accounts, learning topics, interactive scenarios, progress tracking, and recommendation features based on user performance.

2 KIP Product Description

KIP is structured as a web-based educational service built around learning topics, interactive games, user progress data, and source-backed legal content. The system presents users with short misconception-based quizzes and longer branching scenarios, then uses their performance to recommend future practice. Its software components support account access, scenario delivery, feedback generation, content curation, source referencing, and persistent progress tracking. The goal of these features is to assess users' current legal understanding, strengthen their recognition of legal situations, and reinforce informed decision-making through repeated low-stakes practice.

2.1 Key Product Features and Capabilities

KIP combines interactive learning activities, practical legal scenarios, progress tracking, and reliable legal resources to create an engaging legal literacy experience for young adults. These features work together to help users learn legal concepts, apply them to realistic situations,

receive feedback on their decisions, and identify areas where additional practice may be beneficial. The following sections describe the primary features and capabilities that support KIP's educational goals.

2.1.1 Solution Overview

KIP uses focused games, real-world examples, curated media, and official resources to educate young adults on practical legal boundaries.

2.1.2 Goals and Objectives

KIP aims to assess, improve, and reinforce legal knowledge among tech-friendly youth through repeatable, low-stakes interactive activities. Its user experience (UX) is designed to increase recognition of legal situations, improve decision-making skills, and strengthen long-term retention of legal concepts.

2.1.3 Interactive Gamification

KIP uses focused, low-stakes games to improve users' confidence in making legal decisions.

2.1.3.1 "Fact or Fiction?" Game

This short-form game presents users with common legal beliefs and misconceptions, offering "true", "false", or "context-dependent" as answers. This approach will help users challenge assumptions, spot legal nuance, and build foundational knowledge to prepare them for longer and more elaborate scenarios.

2.1.3.2 "Choose Your Outcome" Game

This multiple-choice, long-form game encourages users to interact with people, objects, and documents to learn scenario details presented to them before making a decision. Every choice will demonstrate its consequences to illuminate its legal significance, which encourages users to slow down, evaluate context, and understand what information an outcome is dependent on.

2.1.4 Progress Tracking and Analytics

KIP tracks topical strengths, weaknesses, and completion during use. This information allows KIP to recommend relevant games and topics while helping each game session offer questions, scenarios, and legal concepts that meaningfully test the user.

2.2 Major Components (Hardware/Software)

KIP is composed of several interconnected hardware and software components that work together to provide its web-based educational experience. These components are organized into presentation, application, data, and hardware layers, with each layer supporting a different part of the system. Together, they manage the user interface, application functionality, learning content, data storage, server processing, and access to the platform.

2.2.1 Presentation Layer

The presentation layer provides the user-facing interface for KIP and is responsible for displaying the platform's learning content and interactive features. It is developed using JavaScript, Bootstrap, and HTML5, and includes the account portal, topic browser, game browser, and scenario player.

The account portal allows established users to log in and new users to create an account. Users may also access KIP as guests to explore the platform and play games, although guest users have access to a limited set of features. The topic browser displays available legal learning topics, including traffic law, online conduct, school and work, and personal safety. Users can select a topic to view its available content and track their progress within the topic. The game browser displays KIP's available games and recommends content based on the user's progress, performance history, and demonstrated strengths and weaknesses in specific legal subtopics. Users can also browse and select available games independently. The scenario player serves as the primary interface for interacting with KIP's games. It presents interactive questions, branching scenarios, consequences, feedback, and source references throughout gameplay.

2.2.2 Application Layer

The application layer manages KIP's core functionality and connects the user interface to the platform's data and learning systems. It is developed using Java and JDBC.

2.2.2.1 Authentication System

The authentication system manages user login, account creation, guest access, and session validation. It determines whether a user is using KIP as a registered account or as a limited guest, and it controls which features are available based on that status.

2.2.2.2 Scenario and Feedback Systems

The scenario engine loads and manages the interactive content used during KIP's games. For a play session, it assembles a "match" of play, with multiple "rounds." Each round establishes unique choices, scenario branches, consequences, and game states that appear in sequence. This is primarily informed by two fed selected game and the user's previous actions.

The feedback system provides explanations after user decisions. It identifies whether an answer is correct, incorrect, or context-dependent, explaining the legal reasoning behind the result in order to help the user understand why a specific choice was safe, risky, or incomplete.

2.2.2.3 Curation and Source Reference Systems

The curation system organizes and recommends learning content based on the user's progress and performance history. It determines what games, topics, and scenarios to prioritize, depending on where the user needs more practice, while also allowing users to freely browse and ultimately select any available content.

The source reference system connects game content and feedback to official legal sources, educational resources, and relevant vocabulary. It allows users to see where KIP's information comes from and, when desired, review the original legal language and documents behind a lesson. It also supports content maintenance by tracking changes to legislation, updating references when needed, and flagging KIP content that may be outdated so administrators can review it, revise it, or disable access to it.

2.2.3 Data Layer

The data layer uses SQLite to store and manage KIP's persistent data. It consists of the User Database, Progress Database, Topic Database, Game Database, and Legal Content Database. These databases store user account information, user progress records, learning topics and associated content, game scenario questions and gameplay information, and legal and educational resources.

2.2.4 Hardware Layer

The hardware layer consists of server-side and client-side components that support KIP's operation and accessibility. A virtual private server hosts the Java application, manages server-side processing, and provides access to the SQLite database file and application resources. On the client side, users access KIP through consumer devices with a modern web browser and an internet connection. The hardware layer runs Java application code, stores persistent data, processes user requests, and delivers the web-based interface to users.

3 Glossary

3.1 CDC: Centers for Disease Control and Prevention.

3.2 KIP: Knowledge Is Power.

3.3 MFCD: Major Functional Component Diagram.

3.4 UX: User Experience.

4 References

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